

MCA
(SEM IV) THEORY EXAMINATION 2022-23
ARTIFICIAL INTELLIGENCE

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief. 2 x 7 = 14**

- (a) Discuss MIN-MAX algorithm for game playing.
- (b) Describe the usage of universal and existential quantifier with an example.
- (c) Define Modus Ponens's rule in propositional Calculus.
- (d) Discuss the challenges in reinforcement learning.
- (e) Summarize Support Vector Machine classifier.
- (f) Explain Naïve Bayes classifier.
- (g) Discuss different application areas of AI.

SECTION B**2. Attempt any three of the following: 7 x 3 = 21**

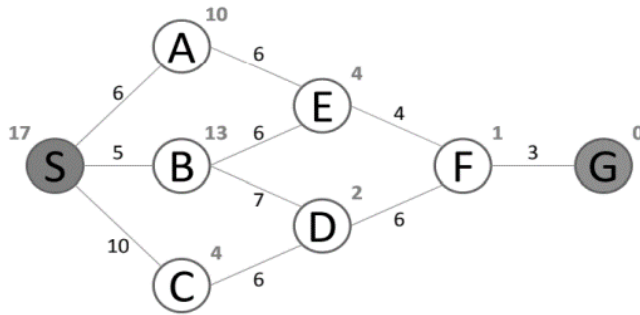
- (a) Discuss possible agent designs for the cases in which clean squares can become dirty and the geography of the environment is unknown. Does it make sense for the agent to learn from its experience in these cases? If so, what should it learn?
- (b) Describe AO* search technique.
- (c) Illustrate Bayesian reasoning. What does a Bayesian network represent? Explain.
- (d) Use forward and backward chaining algorithm for the given knowledge base as:
 $P, P \rightarrow Q, Q \rightarrow R$.
 Derive R by using forward and backward chaining?
- (e) Discuss Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA).

SECTION C**3. Attempt any one part of the following: 7 x 1 = 7**

- (a) Outline the points to describe intelligent agents it is often convenient to specify them in terms of Precepts, Actions, Goals and Environment. Analyze briefly what each of these PAGE concepts mean.
- (b) Analyze the Chinese room argument impractical and how would we would we have to change the Turing test so that it is not subject to this criticism?

4. Attempt any one part of the following: 7 x 1 = 7

- (a) Calculate the A* Algorithm on the following figure. Explicitly write down the queue at each step and find the path till goal state.



- (b) Solve the following Crypt-arithmetic problem
 BASE+ BALL = GAMES

5. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Modify the following sentences to FOPL
- If it is not raining and I have time, then I will go to movie.
 - If it is, raining and I will not go to movie.
 - It is not raining.
 - I will not go to movie.
 - I will not go to movie only if it is not raining.
- (b) Demonstrate the steps for converting FOPL into CNF using an example.

6. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Describe Expectation-Maximization algorithm by choosing a suitable example also discuss its advantage and disadvantage
- (b) Describe the decision tree-learning model by choosing a suitable example. Discuss the issues related to the applications of decision tree

7. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Demonstrate the k-means algorithm, to which category of learning it belongs?
- (b) With the help of a suitable diagram, illustrate the classification process in a pattern recognition system.